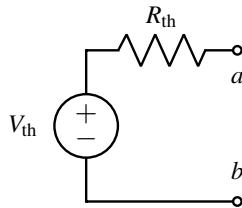


EECS 16B Designing Information Devices and Systems II

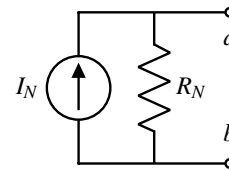
Discussion Worksheet Discussion 2A

1 Circuit Equivalence

To review the circuit equivalence concepts exercised in this worksheet, please see [Note 0B, adapted from the EECS 16A Course Notes](#). We will work through examples in this worksheet. The two forms are presented below for your reference.

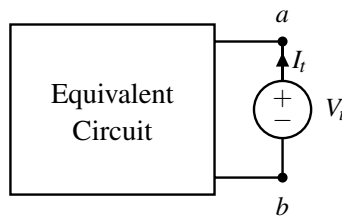


(a) General form of a Thévenin equivalent circuit. Given a circuit and two output terminals, we know the above gives a voltage-source based equivalent; the work lies in solving for V_{th} and R_{th} .

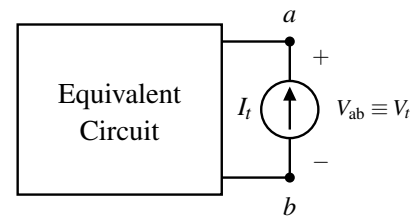


(b) General form of a Norton equivalent circuit, the current-source based equivalent form for a given circuit. Here, we must solve for I_N and R_N .

Below, we display pictorially how to apply current or voltage test sources to a circuit to find R_{eq} .



(a) By applying a test voltage V_t across a and b , we can measure the resulting current draw of the equivalent circuit I_t and use that to calculate R_{eq} .



(b) By feeding a test current I_x into the equivalent circuit, we can measure the resulting voltage drop V_{ab} and calculate R_{eq} .

Figure 2: We can use either a test voltage source or a test current source to find R_{eq} . The choice for what is easier will depend on the specific problem.

2 Transistor Introduction

Transistors (as presented in this course) are 3 terminal, voltage-controlled switches. This means that, when a transistor is "on," the Source (S) and Drain (D) terminals are connected via a low resistance path (short circuit). When a transistor is "off," the Source and Drain terminals are disconnected (open circuit).

Two common types of transistors are NMOS and PMOS transistors. Their states (shorted or open) are determined by comparing the voltage between the G and S terminals ($V_{GS} = V_G - V_S$) to a "threshold voltage"

(V_{tn} for NMOS, V_{tp} for PMOS). Generally, NMOS transistors turn on when V_{GS} is high enough, and PMOS transistors turn on when V_{GS} is low enough (they have complementary behavior!). Transistors are extremely useful in digital logic design since we can use them to implement Boolean logic operators.

In this class, V_{tn} denotes how much **higher** V_G needs to be relative to V_S for the NMOS to be on (allow current flow from drain to source), and $|V_{tp}|$ denotes how much **lower** V_G gate needs to be relative to V_S for the PMOS to be on.

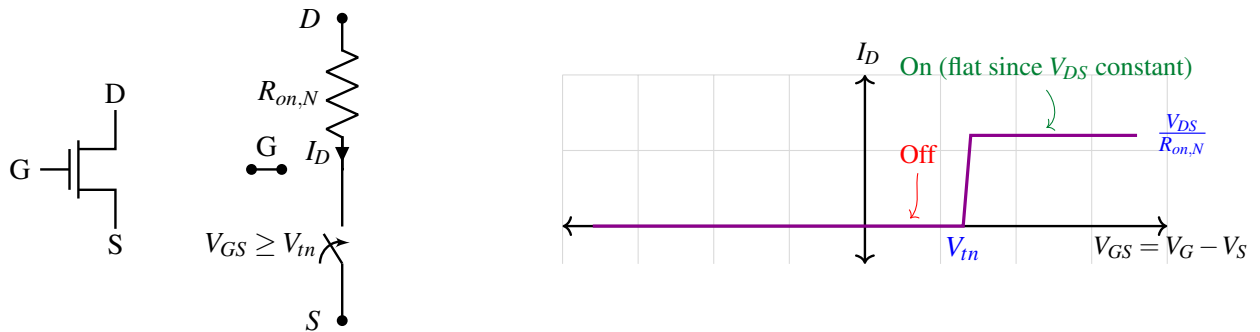


Figure 3: NMOS Transistor Resistor-switch model (the current holding constant at high V_{GS} assumes that V_{DS} is constant).

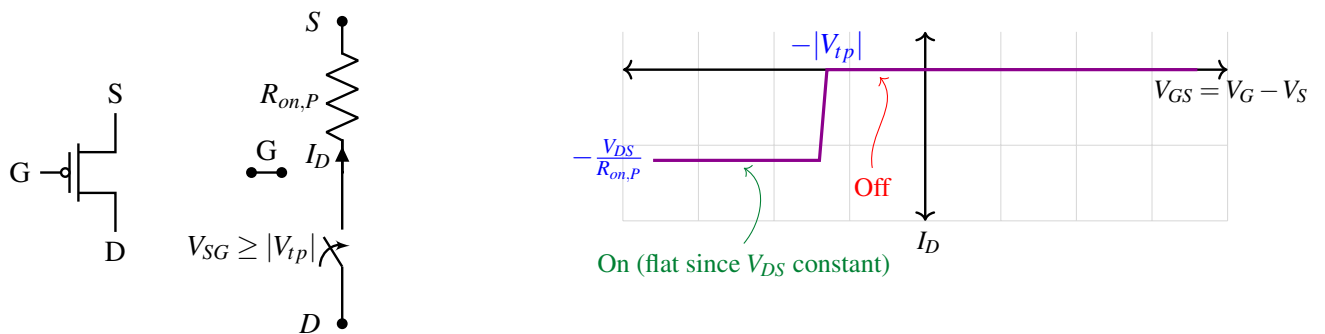


Figure 4: PMOS Transistor Resistor-switch model (the current holding constant at low V_{GS} assumes that V_{DS} is constant). Note that $V_{SG} = -V_{GS}$.

We mentioned that transistors can be connected to perform boolean algebra. An example of this is seen in Section 2, which is called an "inverter" and represents a NOT gate.

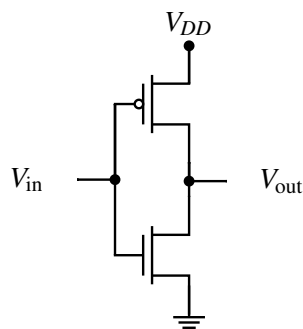


Figure 5: CMOS Inverter

To see why this circuit is called an inverter, we consider the following cases¹:

- (1) **The input is high:** $V_{in} = V_{DD}$. Then, since $V_{GS} \geq V_{tn}$ ($V_{in} \geq V_{tn}$), the NMOS is on. Also, since $V_{GS} \geq V_{tp}$ ($V_{in} - V_{DD} \geq V_{tp} \implies V_{in} \geq V_{DD} - |V_{tp}|$), the PMOS is off. So, only the NMOS switch is closed, and $V_{out} = 0$. That is, **the output is low**.
- (2) **The input is low:** $V_{in} = 0$. Then, since $V_{GS} \leq V_{tn}$ ($V_{in} \leq V_{tn}$), the NMOS is off. Also, since $V_{GS} \leq V_{tp}$ ($V_{in} \leq V_{DD} - |V_{tp}|$), the PMOS is on. So, only the PMOS switch is closed, and $V_{out} = V_{DD}$. That is, **the output is high**.

We can summarize this analysis, using the following truth table:

V_{in}	V_{out}	NMOS	PMOS
V_{DD}	0	on	off
0	V_{DD}	off	on

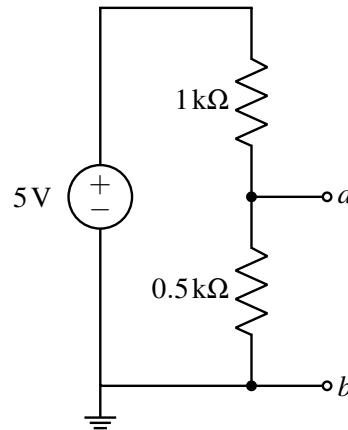
If you think of V_{DD} being a logical 1 and 0V being a logical 0, we have just created the most elementary logical operation using transistors!

¹When working with digital circuits like in Section 2, we usually only consider the values of $V_{in} = 0$ and $V_{in} = V_{DD}$.

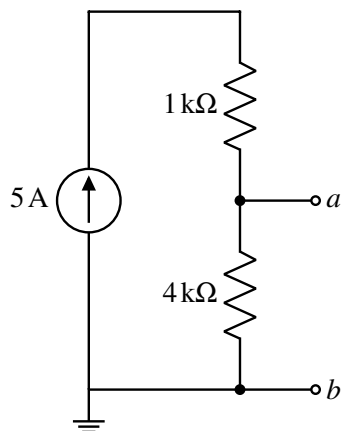
1. Thévenin and Norton Equivalence

Find the Thévenin and Norton equivalents across terminals a and b for the circuits given below. Note that the general forms of these equivalents can be found in Figure 1a and Figure 1b.

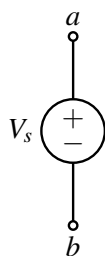
(a)



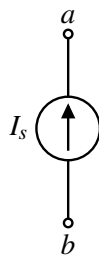
(b)



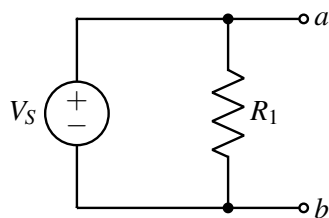
(c) **(Practice)**



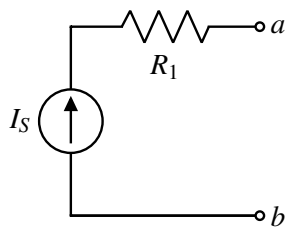
(d) **(Practice)**



(e) **(Practice)**

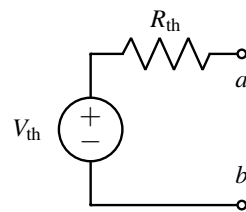
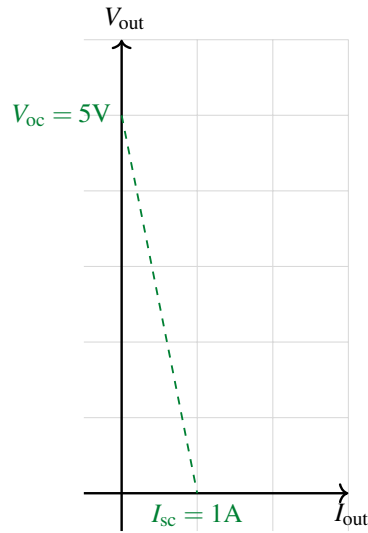


(f) **(Practice)**

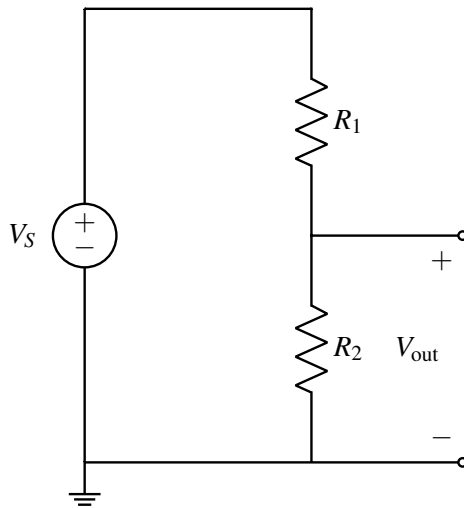


2. Finding Thévenin Equivalents

- (a) You are given the following $I_{\text{out}} - V_{\text{out}}$ characteristic of the Thévenin model of a circuit. Find the Thévenin voltage and the Thévenin resistance. Form a diagram in the style of Figure 1a (copied below for reference).



- (b) You are given a voltage divider as shown below. Find R_1 and R_2 such that the Thévenin equivalent model is the same as that of (a). You are given that $V_S = 10V$.



3. NAND Circuit

Let us consider a NAND logic gate, as seen in Section 2. This circuit implements the boolean function $\overline{(A \cdot B)}$. The $\cdot$ stands for the AND operation, and the $\overline{}$ stands for NOT; combining them, we get NAND!

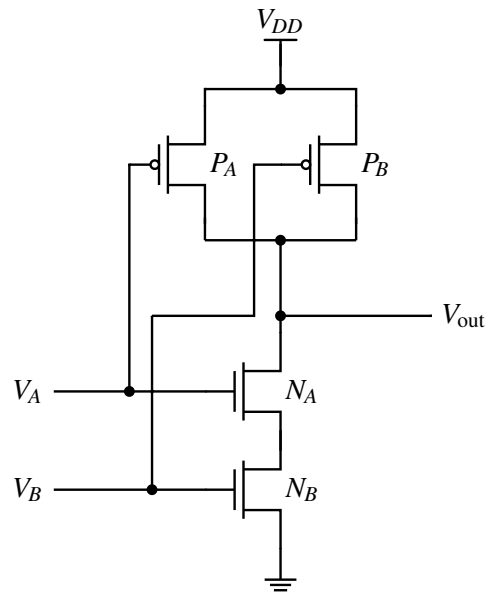


Figure 6: NAND gate transistor-level implementation.

V_{tn} and V_{tp} are the threshold voltages for the NMOS and PMOS transistors, respectively. Assume that $V_{DD} > V_{tn}$ and $|V_{tp}| > 0$.

- (a) Label the gate, source, and drain nodes for the NMOS and PMOS transistors above.

- (b) If $V_A = V_{DD}$ and $V_B = V_{DD}$, which transistors act like open switches? Which transistors act like closed switches? What is V_{out} ?

- (c) If $V_A = 0V$ and $V_B = V_{DD}$, what is V_{out} ?

- (d) If $V_A = V_{DD}$ and $V_B = 0V$, what is V_{out} ?

(e) If $V_A = 0V$ and $V_B = 0V$, what is V_{out} ?

(f) Write out the truth table for this circuit.

V_A	V_B	V_{out}
0	0	
0	V_{DD}	
V_{DD}	0	
V_{DD}	V_{DD}	

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